

Bluestock Mutual Fund Analytics

Final Project Report

End-to-end data engineering, analytics, risk analysis, investor analytics and Power
BI reporting

Capstone Project • September 2026

Executive Summary

The Bluestock Mutual Fund Analytics Capstone is an end-to-end analytics solution that transforms raw mutual fund, market and investor datasets into validated analytical outputs and an interactive Power BI dashboard.

The project combines Python-based ETL, data quality validation, SQLite database design, exploratory data analysis, performance and risk analytics, investor cohort analysis, and a transparent rule-based recommendation model.

The workflow is designed to be reproducible. Project-relative paths are used instead of machine-specific paths, analytical outputs are written to data/processed, and the SQLite database is excluded from Git version control.

Key deliverables include five analytical notebooks, reusable Python scripts, SQL schema and queries, processed datasets, a Power BI dashboard, this final report, and a 12-slide presentation.

Problem Statement & Objectives

Mutual fund datasets contain information across multiple business dimensions: scheme attributes, NAV history, benchmark indices, AUM, SIP activity, portfolios and investor transactions. These sources need consistent preparation before meaningful analysis can be performed.

Objectives:

- Build a reusable ETL pipeline for raw CSV data.
- Establish a relational SQLite database.
- Perform data cleaning and quality checks.
- Explore fund, market, SIP and investor patterns.
- Calculate performance and risk metrics using transparent methodology.
- Measure investor retention through cohort analysis.
- Generate a rule-based analytical recommendation score.
- Deliver an interactive four-page Power BI dashboard.

Data Sources & Data Dictionary

The project uses 10 principal CSV datasets plus selected live NAV outputs. The principal datasets cover fund master information, historical NAV, benchmark indices, scheme performance, AUM, SIP inflows, category inflows, folio counts, portfolio holdings and investor transactions.

fund_master: scheme identifiers, fund house, category, plan, benchmark, expense ratio, loads, minimum investment, manager and risk attributes.

nav_history: AMFI code, date and NAV.

scheme_performance: returns, benchmark return, alpha, beta, Sharpe, Sortino, volatility, drawdown, AUM, expense ratio and ratings.

benchmark_indices: dated index close values.

aum_by_fund_house: fund-house AUM and scheme counts.

monthly_sip_inflows: SIP inflow, active accounts, new accounts, SIP AUM and YoY growth.

category_inflows: monthly category-level net inflow.

industry_folio_count: industry folio counts by segment.

portfolio_holdings: stock, sector, weight, market value and price.

investor_transactions: investor, date, AMFI code, transaction type, amount, location, demographics, payment mode and KYC status.

ETL Architecture & Pipeline

The ETL design follows a simple raw → processed → analytical workflow.

1. Raw ingestion: CSV files are discovered from data/raw using Path-based project resolution.
2. Cleaning: completely empty rows/columns and exact duplicate rows are removed; column names are standardized to lowercase snake_case and text whitespace is trimmed.
3. Processing: cleaned datasets are written to data/processed.
4. Analytics: live NAV retrieval, performance metrics and recommendation generation are executed as separate reusable scripts.
5. Master execution: scripts/run_pipeline.py runs ETL, live NAV retrieval, performance metrics and recommender in sequence.

This design avoids hard-coded absolute paths and supports repeatable execution from the project root.

Data Quality & Cleaning

Data quality checks cover missing values, duplicates, data types, numeric ranges, date parsing and AMFI-code consistency.

The datasets were checked for exact duplicate records. The cleaning workflow removes duplicates without changing legitimate repeated business events.

Meaningful missing observations are retained where the source does not provide a value. In particular, missing SIP YoY growth observations are not blindly imputed.

Negative maximum-drawdown values are retained because drawdown is conventionally represented as a negative percentage. The recommendation model uses the absolute magnitude of drawdown when evaluating risk.

AMFI codes are validated against the fund master so that dependent NAV, transaction, portfolio and performance records can be connected to known schemes.

SQLite Database Design

SQLite provides a lightweight relational layer for the project. The database schema is defined in `sql/schema.sql` and includes `fund_master`, `nav_history`, `aum_by_fund_house`, `monthly_sip_inflows`, `category_inflows`, `industry_folio_count`, `scheme_performance`, `portfolio_holdings`, `investor_transactions` and `benchmark_indices`.

Fund master acts as the reference entity for scheme identifiers. Foreign-key relationships are defined where AMFI codes are shared across datasets.

`scripts/load_database.py` creates and loads the database. `scripts/verify_database.py` checks table structure, row counts and foreign-key relationships.

The generated database is intentionally excluded from GitHub through `.gitignore` because it is a local build artifact.

EDA Findings

Exploratory analysis focuses on fund-house AUM, SIP inflow trends, category-level flows, scheme performance, NAV trends and correlations.

Fund-house analysis provides a view of how assets are distributed across asset-management companies.

SIP analysis highlights changes in monthly contribution activity and supports monitoring of investor participation trends.

Category inflow analysis shows how net flows differ across mutual fund categories.

Scheme-level return analysis enables identification of schemes with higher observed 3-year returns within the supplied dataset.

NAV trend analysis provides a time-series view of selected schemes and establishes the foundation for return-based analytics.

Performance Analytics

Performance analytics are calculated from historical NAV and benchmark data.

CAGR uses actual elapsed years based on the difference between the first and last observation dates. This avoids incorrectly treating every analysis period as exactly 252 trading days.

Annualized volatility is based on daily return standard deviation scaled by the square root of 252.

Maximum drawdown is calculated from the cumulative wealth path and represents the largest peak-to-trough decline.

The resulting performance table is saved as `performance_metrics_calculated.csv` in `data/processed`.

Risk Analytics: VaR, Beta & Sharpe

Historical 95% daily Value at Risk is estimated from the empirical daily-return distribution. The implementation uses the 5th percentile and reports the loss magnitude as a positive percentage.

Beta is calculated by aligning fund daily returns with the selected benchmark return series. NIFTY100 is used when available in the supplied benchmark dataset.

Sharpe ratio is annualized from daily returns. A zero risk-free-rate assumption is used because a separate risk-free-rate series was not supplied.

These metrics should be interpreted as historical analytical measures rather than guarantees of future performance.

Advanced Analytics: Cohort Retention

Cohort analysis groups investors by the month of their first observed transaction.

For each cohort, subsequent activity is measured by months since the cohort's first transaction. The resulting retention matrix allows comparison of continued transaction activity across entry cohorts.

Retention is calculated relative to the cohort's first observed month. A value of 100% in month zero represents the cohort baseline.

The analysis measures transaction activity, not profitability, assets held, or investor satisfaction. A lower observed activity rate therefore should not be interpreted as a direct measure of investment success.

Advanced Analytics: Recommendation Model

The recommender is a transparent rank-based analytical model rather than personalized financial advice.

Weighted components are: 3-year return 30%, Sharpe ratio 25%, volatility 20%, maximum drawdown 15%, and expense ratio 10%.

Higher return and Sharpe ranks receive higher scores. Lower volatility, drawdown magnitude and expense ratio receive higher scores.

The corrected drawdown logic uses the absolute magnitude of the negative drawdown so that a smaller loss magnitude receives a better risk score.

The output is saved as fund_recommendations.csv.

Power BI Dashboard

The Power BI dashboard contains four pages.

Page 1 — Executive Overview: high-level fund count, return and risk indicators, fund-house AUM and top-scheme views, with interactive filters.

Page 2 — Fund Performance & Risk: return, Sharpe, volatility, drawdown and risk/return views, with category and risk filters.

Page 3 — Investor & SIP Analytics: SIP inflows, transaction-type distribution, investor distribution by city tier and YoY SIP growth, with investor filters.

Page 4 — Advanced Analytics: recommendation analysis, VaR, cohort retention and Sharpe-vs-return analysis.

The dashboard is stored as `dashboard/bluestock_mf_capstone.pbix`.

Dashboard Interaction & Business Use

Interactive slicers allow users to move from a broad portfolio view to filtered fund, category or investor segments.

Executive users can inspect fund-house and scheme-level patterns. Analysts can compare risk and return metrics and inspect drawdown and VaR. Investor analytics users can explore transaction activity and SIP behavior.

The dashboard is intended to support exploration and communication of the supplied data rather than replace investment due diligence.

Limitations & Assumptions

Results depend on the supplied datasets, their observation periods and their data quality.

Historical VaR is sensitive to the chosen historical window and does not guarantee future loss estimates.

Sharpe ratio uses a zero risk-free-rate assumption because a risk-free series was not supplied.

Benchmark beta depends on the selected benchmark and date alignment.

Cohort retention is based on observed transaction activity and does not measure profitability.

The recommendation model is rule-based and does not incorporate an individual's goals, liquidity needs, tax situation, or risk profile.

Live NAV retrieval depends on the availability and response of the external MFAPI service.

Recommendations & Future Enhancements

Maintain automated data-quality checks as part of every pipeline execution.

Add scheduled execution using cron or an equivalent scheduler.

Add a Streamlit application for lightweight interactive analysis outside Power BI.

Extend the risk module with Monte Carlo simulation and portfolio optimization where appropriate.

Add automated email delivery of key metrics and data-quality alerts.

Consider publishing the dashboard through Power BI Service or Tableau Public if deployment requirements and account access permit.

Keep methodology and assumptions documented whenever metrics or scoring weights are changed.

Conclusion & Deliverables

The Bluestock Mutual Fund Analytics Capstone establishes a complete workflow from raw data to reproducible analytics and business intelligence reporting.

The solution combines data engineering, database management, EDA, performance measurement, risk analytics, investor cohort analysis and recommendation logic.

Final deliverables: README.md, Python scripts, five notebooks, SQL schema and queries, processed datasets, Power BI dashboard, Final_Report.pdf and Presentation.pptx.

Repository: github.com/farhandev-20/mutual-fund-analytics

Version control excludes local SQLite database artifacts through .gitignore.